1-to'plam. *Amallar tartibi va kasrlarni birlashtiring.*

1. $\frac{3}{4} + 2 \times \left(\frac{1}{2} - \frac{1}{8}\right) =$

2. $\left(\frac{2}{3} + \frac{1}{6}\right) \times 4 - 1 =$

3. $10 - 3 \times \left(\frac{2}{5} + \frac{1}{10}\right) =$

4. $\left(\frac{7}{8} - \frac{1}{4}\right) \div \frac{1}{2} =$

5. $2 \times \frac{3}{4} + \left(1 - \frac{1}{3}\right) \times 6 =$

6. $\left(\frac{1}{2} + \frac{1}{3} + \frac{1}{6}\right) \times 12 - 5 =$

2-to'plam. *Ishorali sonlar va darajalarni birlashtiring.*

1. $(-2)^3 - 3 \times (-4) =$

2. $-3^2 + (-3)^2 =$

3. $(-5)^2 \div (-5) + 2^3 =$

4. $-2 \times (-3)^2 + (-4) =$

5. $(-1)^5 \times (-2)^3 =$

6. $4 - (-2)^4 \div 2 =$

3-to'plam. *O'nli kasr/foiz va tenglamalarni birlashtiring. x ni toping.*

1. $0.25x + 10 = 35$

2. 40% of x is 32

3. x − 15% of x = 68

4. $0.5(x + 4) = 13$

5. 20% of $(x + 10) = 16$

6. $x + 0.1x = 44$

4-to'plam. *Algebraik ifodalar va tenglamalarni birlashtiring. x ni toping.*

1. $3(x - 2) + 5 = 2x + 7$

2. $2(x + 3) - (x - 1) = 10$

3. $4x - (2x + 5) = 7$

4. $5(x - 1) = 3(x + 3)$

5. $-2(x + 4) + 6 = x - 2$

6. $3(2x - 1) - 4 = 2(x + 5)$

5-to'plam. *Kasrli tenglamalarni yeching. x ni toping.*

1. $\frac{2}{3}x - 1 = 5$

2. $\frac{x}{4} + 3 = 7$

3. $\frac{3}{4}x + \frac{1}{2} = 2$

4. $\frac{x+2}{3} = 5$

5. $\frac{2x-1}{5} = 3$

6. $\frac{x}{2} - \frac{x}{3} = 2$

6-to'plam. *$f(x)$ funksiyasi, ishorali sonlar va darajalar.*

1. $f(x) = -2x^2 + 3$. $f(-2) =$

2. $f(x) = x^3 - 4x$. $f(-1) =$

3. $f(x) = (x-2)^2$. $f(-3) =$

4. $f(x) = 2x^2 - 5x + 1$. $f(-1) =$

5. $f(x) = -x^2 + 2x$. $f(3) =$? $f(-3) =$

6. $f(x) = kx + b$. $f(2) = 7, f(0) = 3$. $k =$? $b =$

7-to'plam. *Xatoni toping – ikki mavzu aralashgan, bitta xato qadam bor.*

1 $\frac{1}{2}x + 3 = 7$ ifodasida ko'rsatilgan yechimda xato bor:

"$\frac{1}{2}x = 7 + 3 = 10$, *shuning uchun* $x = 20$."

Xato va to'g'ri javob: .....

2 $3(x - 2) = 9$ ifodasida ko'rsatilgan yechimda xato bor:

"$3x - 2 = 9$, *shuning uchun* $3x = 11$, $x = \frac{11}{3}$."

Xato va to'g'ri javob: .....

3 $f(x) = x^2 - 2x$, $f(-3)$ ifodasida ko'rsatilgan yechimda xato bor:

"$f(-3) = (-3)^2 - 2(-3) = 9 - 6 = 3$."

Xato va to'g'ri javob: .....

4 $0.4x = 24$ ifodasida ko'rsatilgan yechimda xato bor:

"$x = 24 \times 0.4 = 9.6$."

Xato va to'g'ri javob: .....

5 $2(x + 1) - x = 9$ ifodasida ko'rsatilgan yechimda xato bor:

"$2x + 1 - x = 9$, $x + 1 = 9$, $x = 8$."

Xato va to'g'ri javob: .....

8-to'plam. *Bonus / eng qiyin – bitta masalada 3+ ko'nikma.*

1 $\frac{1}{2}(x - 4) + 0.25x = 5$

2 $f(x) = -2(x - 1)^2 + 3$. $f(-1) =$

3 30% of $(2x + 10) = x + 2$

4 $\frac{3}{4}x - (-2)^3 = 11$

5 $f(x) = kx + b$. $f(-2) = 1$, $f(2) = 9$. $k =$? $b =$? $f(0) =$

9-to'plam. *SAT'da qanday uchraydi – 1–8-darslarning barchasidan.*

1 $3 + 2 \times (7 - 4)^2$ ifodaning qiymati nechaga teng?

(A) 15 (B) 21

(C) 30 (D) 45

2 Retseptda $\frac{3}{4}$ stakan shakar kerak. Retseptning $\frac{2}{3}$ qismini tayyorlash uchun necha stakan shakar kerak?

(A) $\frac{5}{12}$ (B) $\frac{1}{2}$

(C) $\frac{3}{4}$ (D) $\frac{2}{3}$

3 Ko'ylak \$40 turadi va 25% chegirma bilan sotilmoqda. Chegirmali narx nechaga teng?

(A) \$10 (B) \$15

(C) \$30 (D) \$35

4 $(-4) + (-3) \times (-2)$ ifodaning qiymati nechaga teng?

(A) -10 (B) -2

(C) 2 (D) 14

5 $3(2x - 1) - (x + 4)$ ifodasiga teng ifodani toping.

(A) $5x - 7$ (B) $5x + 7$

(C) $7x - 7$ (D) $2x - 7$

6 $4x - 5 = 2x + 9$ tenglamasini qanoatlantiruvchi x qiymatini toping.

(A) 2 (B) 5

(C) 7 (D) 9

7 $2^3 \times 2^2$ ifodaning qiymati nechaga teng?

(A) 10 (B) 16

(C) 32 (D) 64

8 Agar $f(x) = 2x - 5$ bo'lsa, $f(4)$ nechaga teng?

(A) -1 (B) 3

(C) 8 (D) 13

9 $\frac{1}{2} + \frac{3}{4} \times 2$ ifodaning qiymati nechaga teng?

(A) 1 (B) $\frac{5}{4}$

(C) $\frac{3}{2}$ (D) 2

10 -20 va 16 sonlari yig'indisining 50%-i nechaga teng?

(A) -18 (B) -2

(C) 2 (D) 18

11 $x > 0$ bo'lsa, $x^2 = 49$ tenglamasini qanoatlantiruvchi x nechaga teng?

(A) 7 (B) -7

(C) 24.5 (D) 98

12 Agar $f(x) = -x^2 + 3$ bo'lsa, $f(-2)$ nechaga teng?

(A) -7 (B) -1

(C) 1 (D) 7

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	$\frac{3}{2}$	2	$\frac{7}{3}$
	3	$\frac{17}{2}$	4	$\frac{5}{4}$
	5	$\frac{11}{2}$	6	7
2-to'plam	1	4	2	0
	3	3	4	-22
	5	8	6	-4
3-to'plam	1	$x=100$	2	$x=80$
	3	$x=80$	4	$x=22$
	5	$x=70$	6	$x=40$
4-to'plam	1	$x=8$	2	$x=3$
	3	$x=6$	4	$x=7$
	5	$x=0$	6	$x=\frac{17}{4}$
5-to'plam	1	$x=9$	2	$x=16$
	3	$x=2$	4	$x=13$
	5	$x=8$	6	$x=12$
6-to'plam	1	-5	2	3
	3	25	4	8
	5	$f(3)=-3$, $f(-3)=-15$		
	6	$k=2$, $b=3$		
7-to'plam	1. Teskari amal xato – 3 qo'shish o'rniga ayirish kerak edi. To'g'ri: $\frac{1}{2}x = 4$, $x = 8$.			
	2. 3 faqat x ga taqsimlangan, -2 ga emas. To'g'ri: $3x - 6 = 9$, $x = 5$.			
	3. $-2 \times (-3) = -6$ emas, $+6$. To'g'ri: $9 + 6 = 15$.			
	4. Ko'paytirish o'rniga bo'lish kerak edi. To'g'ri: $x = 24 \div 0.4 = 60$.			
	5. 2 faqat x ga taqsimlangan, $+1$ ga emas. To'g'ri: $2x + 2 - x = 9$, $x = 7$.			
8-to'plam	1	$x=\frac{28}{3}$		
	2	-5	3	$x=\frac{5}{2}$
	4	$x=4$	5	$k=2, b=5$
9-to'plam	1	B	2	B
	3	C	4	C
	5	A	6	C
	7	B	8	B
	9	D	10	B
	11	A	12	B

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

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